Mini Project A – Checklist and Marking Rubric

Stage A1: 5% of course grade | Stage A2 + Report A: 10% of course grade

Assessment Structure

Assessment component	Maximum marks	Contribution to course grade
Stage A1 - SUMO network development	100	5%
Stage A2 - simulation and intervention	60	Part of combined 10%
Report A - research paper (Stages A1 + A2)	40	Part of combined 10%
Stage A2 + Report A	100	10%

Fixed-Score Performance Scale

Marking policy: Each criterion receives one fixed score: 100%, 90%, 75%, 60%, 35%, or 0% of its allocated marks. We would not assign marks between these levels.

Performance level	Fixed score	General interpretation
Meets Requirements	**100%**	Complete, technically correct, and supported by appropriate evidence.
Needs Limited Improvement	**90%**	Substantially complete and correct; only small corrections or additions are needed.
Needs Moderate Improvement	**75%**	Core requirement is met, but several issues remain.
Needs Major Improvement	**60%**	Important elements are present, but significant errors weaken the work.
Needs Substantial Improvement	**35%**	Meaningful attempt is evident, but major deficiencies prevent the criterion from being adequately met.
Not Completed / No Meaningful Evidence	**0%**	Criterion is missing, not attempted, or lacks enough meaningful evidence to assess.

Stage A1 - Network Development

Maximum: 100 marks | Weight: 5% of final course grade

Milestone / criterion	Marks	Full-credit standard (100%)
Milestone 1: Create a new network and import GIS imagery	**20**	Creates the SUMO project and imports the assigned GIS imagery with appropriate coordinates, scale, and alignment.
Milestone 2: Create a road network on the GIS map	**45**	Represents the visible major, secondary, and local roads throughout the assigned study area with appropriate geometry, lane counts, roadway details, and reasonable correspondence to the GIS imagery.
Milestone 3: Junction modeling fundamentals	**35**	Models junctions, turning lanes, turning movements, and lane-to-lane connections consistently throughout the network so intended vehicle movements can occur without major connectivity conflicts.

Stage A2 - ITS and Intervention

Maximum: 60 marks | Combined with Report A for 10% of final course grade

Scope: Assign traffic demand and applicable traffic-signal timing on the assigned intersection only; develop 100% HDV, 100% CAV, and 100% HDV + ONE intervention scenarios. Analyze the Speed-Density relationship on ONE selected major-road edge. Evaluate the intervention on that same edge.

We have provided each group with a PowerPoint presentation containing detailed instructions for their assigned intersection, including traffic volumes, traffic signal settings, and other relevant information.

Milestone / criterion	Marks	Full-credit standard (100%)
Milestone 1: Traffic flow	**10**	Configures traffic demand on major roads of the intersections with appropriate routes, arrival assumptions, and documented inputs; simulation traffic moves as intended.
Milestone 2: Traffic signals	**10**	Implements applicable traffic signals at required major-road intersection; signal phases and timings support the intended movements and are documented.
Milestone 3: Adjust right-of-way	**8**	Configures junction priorities and right-of-way rules consistently with intended network movements and avoids major priority conflicts.
Milestone 4: Simulate HDVs and CAVs	**16**	Runs documented 100% HDV and 100% CAV scenarios using consistent network, demand, routes, signals, and simulation duration; only intended vehicle parameters/composition differ.
Milestone 5: One intervention	**16**	Selects and justifies ONE intervention, documents before/after settings, runs the third 100% HDV scenario, and evaluates its effects on the SAME selected edge using Speed-Density evidence.

Report A - Research Paper (Stages A1 and A2)

Required Preparation for Report A - Generative AI Writing Videos

Before beginning the Report A writing process, every group should watch both videos below. The videos demonstrate a workflow for using generative AI to support research-paper/technical-report writing.

Video 1: https://youtu.be/DsGvDcmw5ds

Video 2: https://youtu.be/21tJPYt951s

After watching the vidoes, download **the below provided word file in Avenue** and start writing:

1. WritingStructure.docx
2. Template.docx
3. GroupX.pptx (for Figures samples)

Note: Generative AI might be used only as a starting point for drafting, and it may not fully follow all the instructions below. I highly recommend combining AI-generated content with your own effort to review, revise, verify, and significantly improve the final report.

Maximum: 40 marks | Assesses documentation, analysis, and communication of the integrated investigation

Criterion	Marks	Full-credit standard (100%)
Research-paper structure **(See WritingStructure.docx)**	8	Uses the prescribed title/abstract and five main sections in order; aligns the three project problems with objectives, literature review, methodology, results, and conclusions; covers BOTH Stage A1 and Stage A2.
Engineering analysis and interpretation **(See Result Analysis and conclusion sub sections in WritingStructure.docx)**	12	Documents the Stage A1 network and Stage A2 assumptions; compares HDV and AV Speed-Density results; justifies and evaluates ONE intervention using before/after evidence for the SAME selected edge; conclusions address all three objectives.
Formatting **(See sub-section formatting in Template.docx)**	4	Consistently follows the required report format: Times New Roman 11 pt body text, 1.5 line spacing, justified paragraphs, single-column layout, 1-inch margins, and the specified title/heading styles. Uses capitalization and abbreviations consistently, and numbers, captions, and references all figures and tables appropriately. Keywords, where required, are concise and derived from the abstract.
Writing style and clarity **(See sub-section Writing style and clarity in Template.docx)**	7	Presents clear, concise, and technically accurate engineering writing. Uses active voice and appropriate tense, consistent terminology for the same concepts, direct sentence structure, and logical transitions. Avoids unnecessary adverbs, overly technical language, long or indirect clauses, and unnecessary words. Defines technical terms when needed and communicates the analysis in a precise and readable manner.
Referencing **(See sub-section Referencing in Template.docx)**	3	Uses relevant and verifiable sources with consistent APA-style referencing. Provides accurate author–year in-text citations, correctly uses et al. for three or more authors, appropriately cites multiple sources, and includes a complete and consistently formatted APA reference list for academic papers, reports, and other sources used in the report.
Figures **(See GroupX.pptx)**	6	Presents clear, high-resolution figures without blurriness or distortion. All figures are appropriately numbered, captioned, and directly referenced and explained in the report text. Uses readable labels, legends, and units. For multi-panel figures (a, b, c, d, etc.), maintains consistent panel widths, heights, alignment, and formatting.

First complete the technical work and analysis, then write the report based on your actual results

Now, please read the SMRTTECH-4SC3-MiniProjectsA.docx for the detailed project description.